

AN14875

SR250 hardware design guide

Rev. 1.0 — 27 March 2026

Application note

Document information

Information	Content
Keywords	SR250 hardware design, power modes, IC interfaces
Abstract	This documents explains how to integrate the SR250 UWB RF transceiver, covering power interfaces, SPI communication, clock configurations, RF architecture, and PCB layout guidelines.



1 Introduction

This document explains how to integrate the SR250 UWB RF transceiver, covering power interfaces, SPI communication, clock configurations, RF architecture, and PCB layout guidelines.

The SR250 is a fully integrated single chip Impulse Radio Ultra-Wideband (IR-UWB) low-energy low-cost transceiver IC of the second generation.

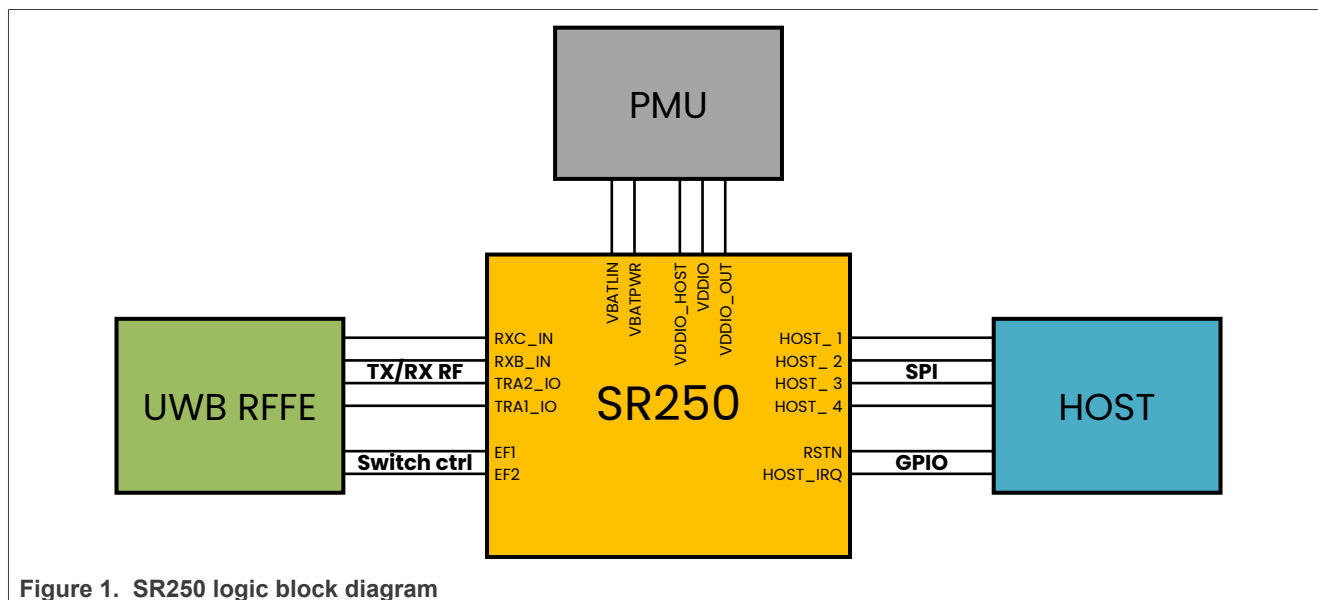
Main features:

- Supports UWB bands channel 5 and 9 for worldwide use.
- Optimized for short frame mode operation at maximum FCC TX limits.
- Programmable transmitter output power
- Conforms to the regulatory requirements of FCC, ARIB & ETSI UWB
- Low power consumption
- Supports 2-way ranging and one-way ranging TOA measurement.
- Supports Angle of Arrival (AoA) measurement
- Support radar with on-chip presence detection.
- ARM® Cortex-M33 32-bit Processor at 125MHz

As shown in [Figure 1](#) below, SR250 requires the following connections to operate:

- PMU connections for power supply – it could be direct battery supply.
- HOST connections – ranging/radar results control and reading
- UWB RF front-end connection – for antenna connection and RF switch control

All connections and their configurations are presented in the following sections.



2 Typical application schematics

The following sections present the main application schematics for the SR250.

2.1 SR250 reference schematic – Default configuration

Figure 2 shows the reference schematic when SR250 DC-DC is in use. VBAT is in the range between 2.3 V and 5 V. The DC-DC converter steps down the voltage to supply the internal LDO with high yield. For more information, refer to [Section 3.1](#).

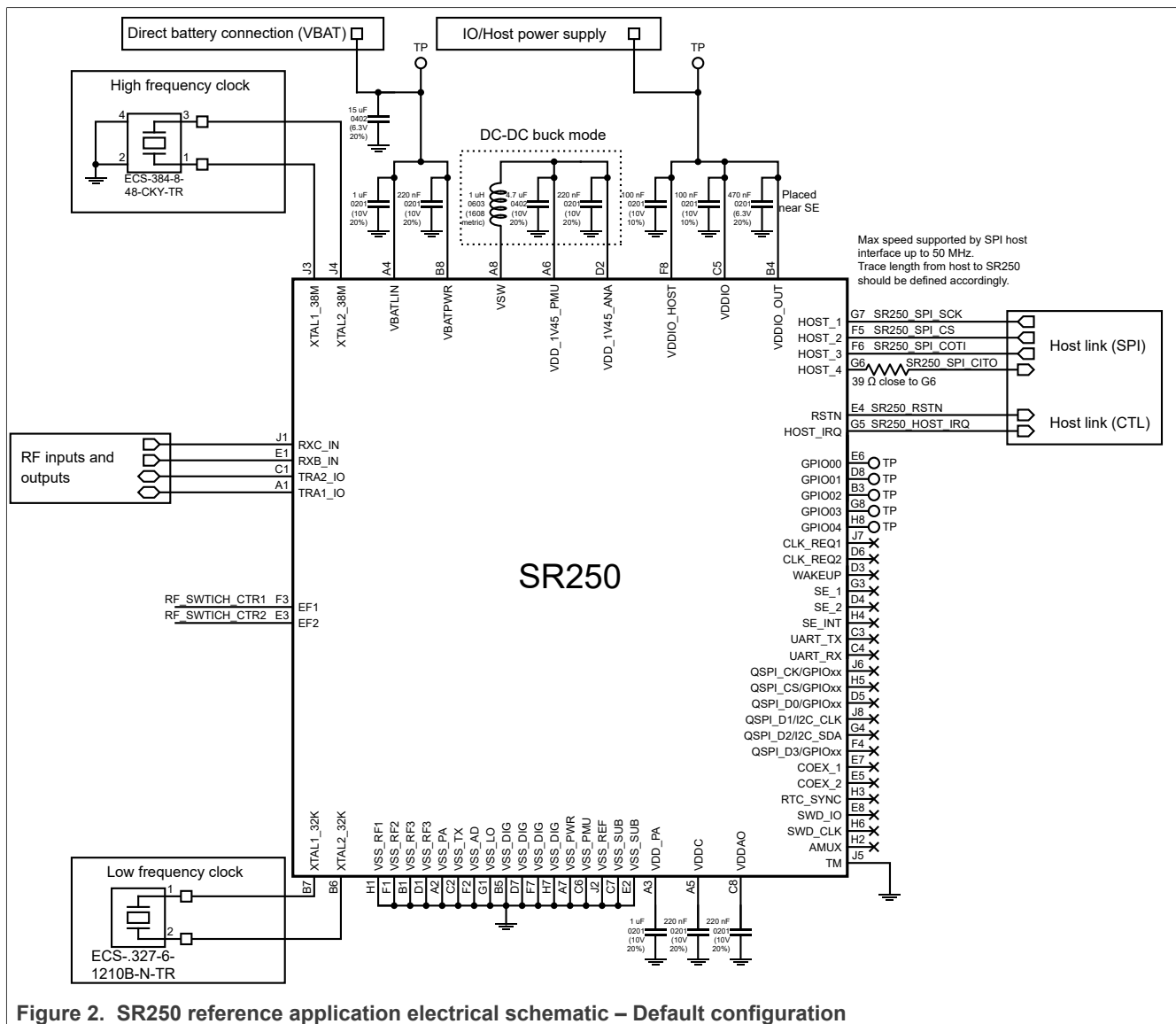


Figure 2. SR250 reference application electrical schematic – Default configuration

2.2 SR250 reference schematic – BOM optimized

In case of 1.8 V robust and low noise LDO is available on the platform and/or if coil integration is a constraint, SR250 DC-DC can be bypassed. In this case, SR250 can be supplied directly by a 1.8 V power rail. However, with this configuration, application designs cannot benefit from the DC-DC power transfer yield. The maximum

TX power is approximately 2.5 dB lower than the default configuration shown in [Section 2.1](#) (refer to the data sheet).

[Figure 3](#) shows the reference schematic when SR250 is powered by a 1.8V power rail. For more information, refer to [Section 3.2](#).

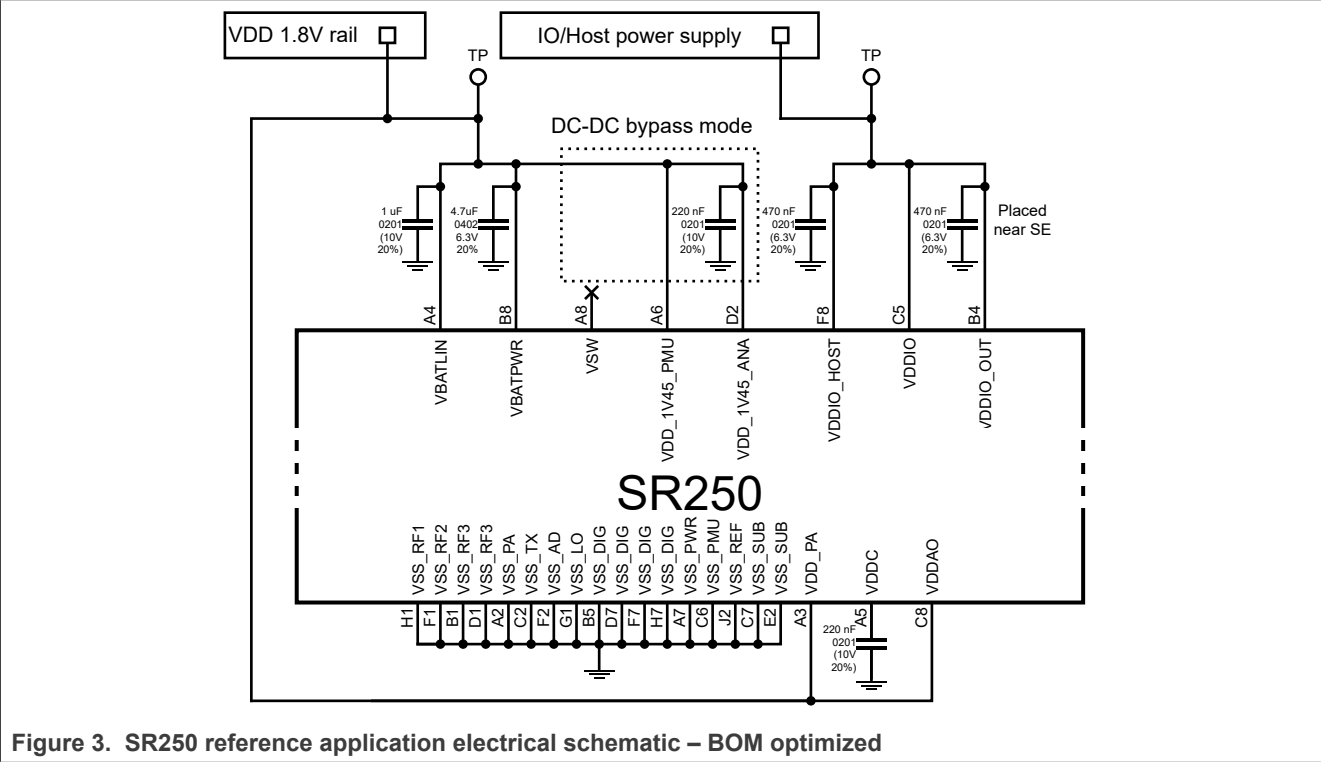


Figure 3. SR250 reference application electrical schematic – BOM optimized

3 Power interface

3.1 Default configuration

3.1.1 External power supplies

The SR250 requires several external power supplies, split into different power domains:

- Main power supply – battery powered:
 - VBATPWR: DC-DC buck supply
 - VBATLIN: Power amplifier LDO supply
- IO power supply:
 - VDDIO: IO pads supply voltage for GPIO
 - VDDIO_HOST: IO pads supply voltage for UWB Host

Table 1. External power supplies voltages

Pin	Conditions	Min.	Typ.	Max.	Units
VBATPWR	-	2.3	3.6	5	V
VBATLIN	-	2.3	3.6	5	V
VDDIO	1.2 V	1.08	1.2	1.32	V
VDDIO_HOST		1.08	1.2	1.32	V
VDDIO	1.8 V	1.62	1.8	1.98	V
VDDIO_HOST		1.62	1.8	1.98	V

3.1.2 Decoupling capacitors (default configuration)

Note: *Component de-rating over voltage and temperature must be carefully considered during the decoupling capacitors selection process.*

The recommended external capacitors are listed below:

Table 2. Decoupling capacitors (default configuration)

Group	Name	Voltage	Nom. values	Tol. [%]
Main power	VBATPWR	10 V	220 nF	±20
		6.3 V	15 µF	±20
	VBATLIN	10 V	1 µF	±20
DC-DC buck	VDD_1V45_PMU	10 V	4.7 µF	±20
	VDD_1V45_ANA	10 V	220 nF	±20
Internal LDO	VDD_PA	10 V	1 µF	±20
	VDDC	10 V	220 nF	±20
	VDDAO	10 V	220 nF	±20
IO	VDDIO	10 V	100 nF	±10
	VDDIO_HOST	10 V	100 nF	±10
	VDDIO_OUT	6.3 V	470 nF	±20

3.2 BOM optimized configuration

3.2.1 External power supplies

In this configuration, the SR250 requires two separate 1.8 V external power supplies:

- Main 1.8 V power rail, which connects VBATLIN, VBATPWR, VDD_1V45_ANA, VDD_1V45_PMU, VDD_PA and VDD_AO.
- IO 1.8 V domain, connecting VDDIO and VDDIO_HOST.

Table 3. External power supplies voltages

Pin	Conditions	Min.	Typ.	Max.	Units
VBATPWR	-	1.62	1.8	1.98	V
VBATLIN	-	1.62	1.8	1.98	V
VDDIO	1.2 V	1.08	1.2	1.32	V
VDDIO_HOST		1.08	1.2	1.32	V
VDDIO	1.8 V	1.62	1.8	1.98	V
VDDIO_HOST		1.62	1.8	1.98	V

3.2.2 Decoupling capacitors (BOM optimized configuration)

Note: *Component de-rating over voltage and temperature must be carefully considered during the decoupling capacitors selection process.*

The recommended external capacitors are listed below:

Table 4. Decoupling capacitors (BOM optimized configuration)

Group	Name	Voltage	Nom. values	Tol. [%]
Main power	VBATPWR	6.3 V	4.7 μ F	± 20
	VBATLIN	10 V	1 μ F	± 20
DC-DC bypass	VDD_1V45_PMU	-	-	-
	VDD_1V45_ANA	10 V	220 nF	± 20
Internal LDO	VDD_PA	-	-	-
	VDDC	10 V	220 nF	± 20
	VDDAO	-	-	-
IO	VDDIO	-	-	-
	VDDIO_HOST	6.3 V	470 nF	± 20
	VDDIO_OUT	6.3 V	470 nF	± 20

4 Host interface (SPI)

The following connections are used for communication between the host and the SR250 via SPI bus. These pins cannot be left open. The required pins and functionalities are described in [Table 5](#). Host connection pads are supplied by VDDIO_HOST.

Table 5. Host connections

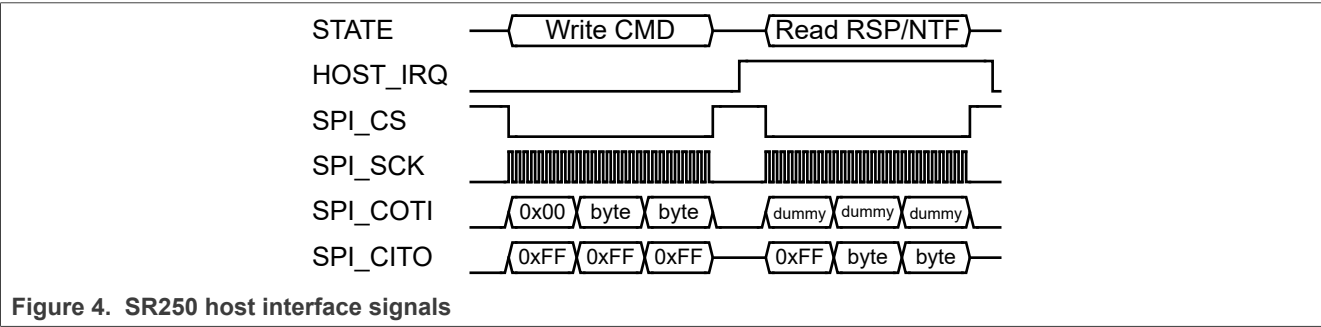
Pin name	Functionality
HOST_1	SCK (Serial Input Clock) connected to the host (SPI)
HOST_2	CS (Not Target Select) connected to the host (SPI)
HOST_3	COTI (Controller Out Target In) - data input connected to the host (SPI)
HOST_4	CITO (Controller In Target Out) - data output connected to the host (SPI) A 39 Ω serial resistor is required – to be placed close to the SR250 G6 pin (see Figure 2) to minimize undershoot and overshoot.
HOST_IRQ	IRQ sent to the HOST to let it know data are available for read
RSTN	Reset pin

Host GPIOs are directly connected to the GPIOs of the application processor (no additional components needed).

In SR250, SPI is adopted with a direction byte feature, on every CS assertion will be followed by a direction byte. This direction byte is controlled purely by the host. This means the host has control over the SPI bus to perform read/write operations.

Table 6. SPI direction bytes decoding

[7:6] bits of 1-byte	Description
00/01	Write from controller to target
11	Read from target
10	Invalid access from controller



5 CLK interface (XTAL)

The SR250 chip requires two separate clocks:

- High frequency clock (38.4 MHz/26 MHz): Used as clock reference for the two PLLs (RFPLL and SYSPLL) and for maintaining the time grid whenever the SR250 is not in DPD.
- Low frequency clock (32.768 kHz): Used as reference clock for RTC (real time clock). RTC is used for maintaining the time grid when the SR250 is in DPD.

Internally, the SR250 default trimming is performed so that no additional load capacitors are required on the PCB for XTAL if XTAL load capacitance is within the specified range (refer to data sheet).

Additional load SMD capacitors are **not recommended** by NXP.

5.1 High frequency clock XTAL

The recommended XTAL load capacitance for the high frequency clock is 8 pF. Otherwise, it should stay within the limits defined in the SR250 datasheet (6 pF to 10 pF)

- If the XTAL load capacitance > 6 pF **but** not 8 pF then trimming capacitors should be added for compensation. The UCI parameter for trimming capacitance control is called 38.4MHz_XTAL_CAP_GM_CTRL (refer to [AN13848 – SR250 calibration procedure guide](#) and [UM12508 – SR250 UCI specification](#)).
- NXP does not recommend using XTAL load capacitances lower than 6 pF as they can lead to errors/shifts in frequency.

Find a list of suitable XTAL references below:

Table 7. XTAL references for high frequency clock

Reference	Vendor	Load Capacitance	Frequency tolerance
SXT1148FE48-38.400M	SUNTSU Electronics	8 pF	+/-10 ppm
ECS-384-8-48-CKY	ECS	8 pF	+/-10 ppm
CX1210DB38400 D0FLFC4	KYOCERA	8 pF	+/-10 ppm

5.2 Low frequency clock XTAL

The recommended XTAL load capacitance for the low frequency clock is 7 pF. Otherwise, it should stay within the limits defined in the SR250 datasheet [7 pF to 12.5 pF].

- NXP does not recommend using XTAL load capacitances lower than 7 pF as they can lead to errors/shifts in frequency.

Find a list of suitable XTAL references below:

Table 8. XTAL references for low frequency clock

Reference	Vendor	Load Capacitance	Frequency tolerance
ECS-.327-7-1210-TR	ECS	7 pF	+/-20 ppm

6 RF architecture

6.1 RF architecture: UWB ranging and radar

The SR250 has three RX chains and one TX chain. In terms of IC pins, two pins are dedicated to the receiver (RXB, RXC) and two pins (TRA1, TRA2) can be used as transmitter or receiver. With the SR250 chip, no additional RF switch is needed on the PCB to support up to four antennas.

In the following section, RF architectures are proposed depending on the UWB feature targeted. From this optimized baseline, any configuration can be derived to address specific needs (for example, five antenna designs, single or dual antenna use cases, etc.).

6.1.1 UWB ranging ToF

6.1.1.1 Ranging basic ToF

In case only ranging is needed for the application, the user can choose to design the SR250 with a single antenna only.

- For single antenna implementation unused RF pins are left OPEN.
- The reference schematic below shows the antenna path for single antenna implementation. It is recommended to use TRA2 for the RF port.

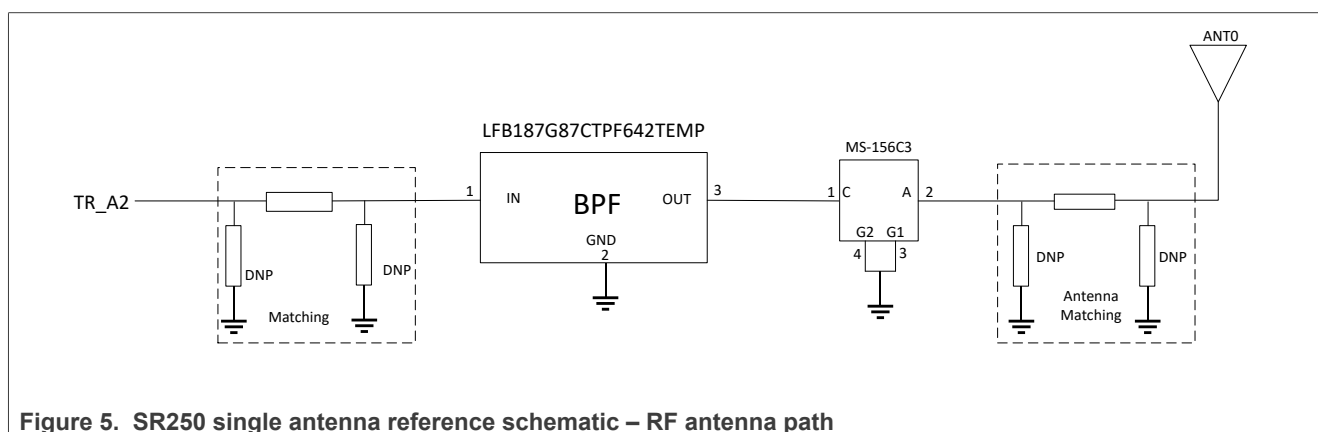


Figure 5. SR250 single antenna reference schematic – RF antenna path

6.1.1.2 Ranging ToF with antenna diversity

Based on the use case, an IOT device would need to range efficiently in a defined pattern. It is then recommended to use multiple antennas to avoid having one antenna radiation pattern blocked by an obstacle (for example, a human or an object).

The SR250 offers two independent TX/RX pins. It is recommended to connect these pins to the antenna in such a way that they can still radiate despite potential blocking.

In addition, reception sensitivity is improved when using additional receivers (better sensitivity due to maximal ratio combining (MRC) gain). Therefore, to achieve optimum ranging performances, RXB and RXC can be connected to separate antennas should be left open if not used).

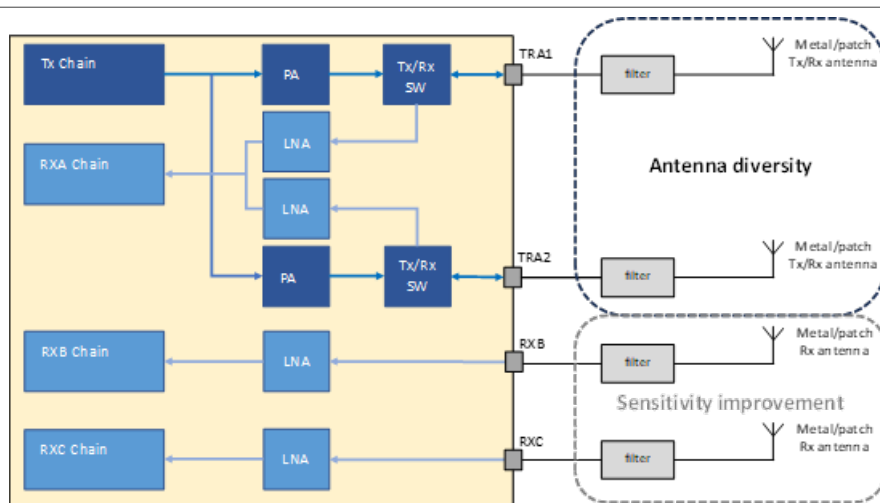


Figure 6. SR250 RF architecture optimized for UWB ranging ToF

6.1.2 UWB ranging AoA

Ranging AoA (2D and 3D) can be achieved by using patch antennas separated by a distance smaller than $\lambda/2$ (where λ corresponds to the channel wavelength of the highest frequency for example, CH9~8GHz) and arranged in an L-shape configuration (for 3D).

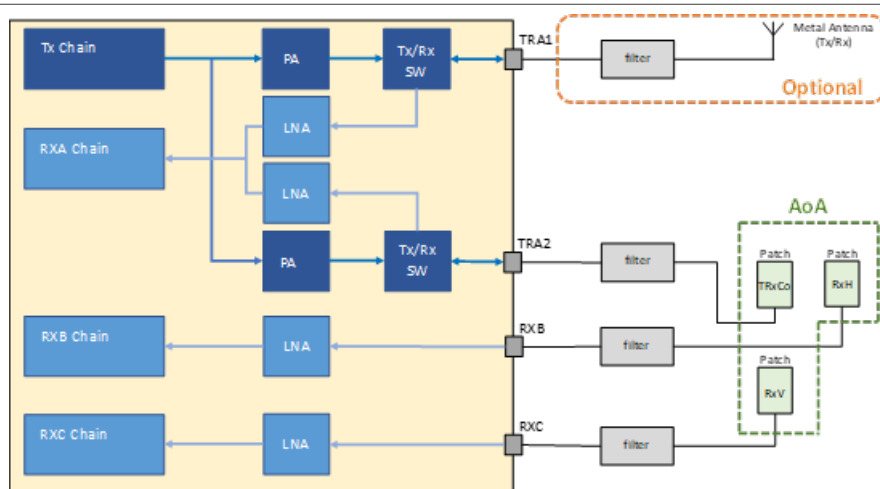


Figure 7. SR250 RF architecture optimized for ranging AoA

The schematic shown in [Figure 7](#) is optimized for 3D AoA which is the combination of a horizontal patch antenna pair (TRA2, RXB) and a vertical patch antenna pair (TRA2, RXC). A common patch antenna between H and V pairs is also used for TX.

Optionally, a second TX/RX antenna can be added to mitigate signal blocking and guarantee ranging.

6.1.3 UWB radar ToF

For UWB radar ToF (used for radar detection), one dedicated TX antenna (metal or a patch antenna) and one dedicated RX patch antenna are required as radar mode transmission and reception are performed simultaneously.

It is recommended to use TRA1 and RXB as the RX sensitivity is slightly better compared to TRA2 and RXC. Unused RF ports are left open.

Except for some specific use cases, using a patch antenna for RX is recommended as the antenna FoV is limited to the front side of the patch.

Whatever the TX/RX pair used, TX-RX isolation (see [Section 6.1.6](#)) is a key factor that determines the performances of the setup (performances in terms of maximum distance detection).

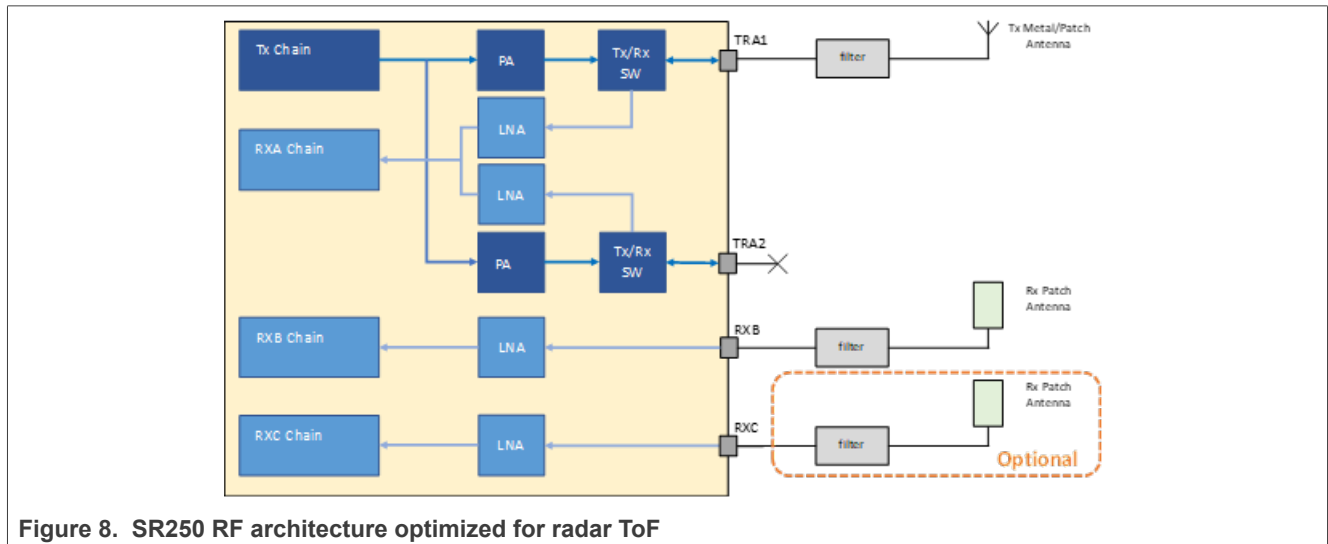


Figure 8. SR250 RF architecture optimized for radar ToF

For better radar Rx sensitivity, a second RX antenna connected to RXC can be added.

6.1.4 UWB radar detection AoA

Same as for UWB ranging AoA, UWB radar 2D AoA requires patch antenna pairs (horizontal) separated by a distance smaller than $\lambda/2$ (where λ corresponds to the channel wavelength of the highest frequency for example, CH9~8GHz).

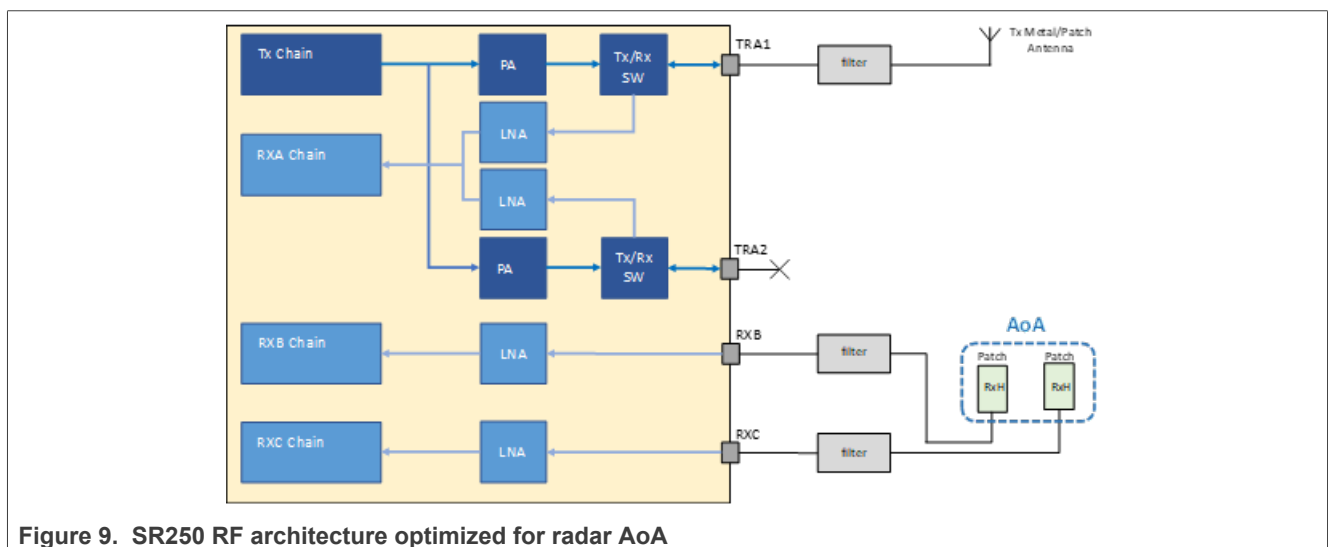


Figure 9. SR250 RF architecture optimized for radar AoA

6.1.5 Optimized architecture for UWB ranging and radar detection (ToF/AoA)

Figure 10 shows an RF architecture optimized for UWB ranging and radar detection. This architecture can support all the previously mentioned scenarios:

- Ranging ToF antenna diversity is using two antennas (metal + patch) connected to the two TX/RX ports.
- Ranging AoA (3D and 2D) is achieved using two/three patch antennas connected to two/three RX ports.
- Radar detection ToF is improved by the usage of a metal antenna (or a patch antenna) to transmit and **two** patch antennas to receive simultaneously (with improved sensitivity).
- Radar detection AoA (2D) is using the independent metal antenna for TX and one horizontal patch antenna pair connected to two RX ports.

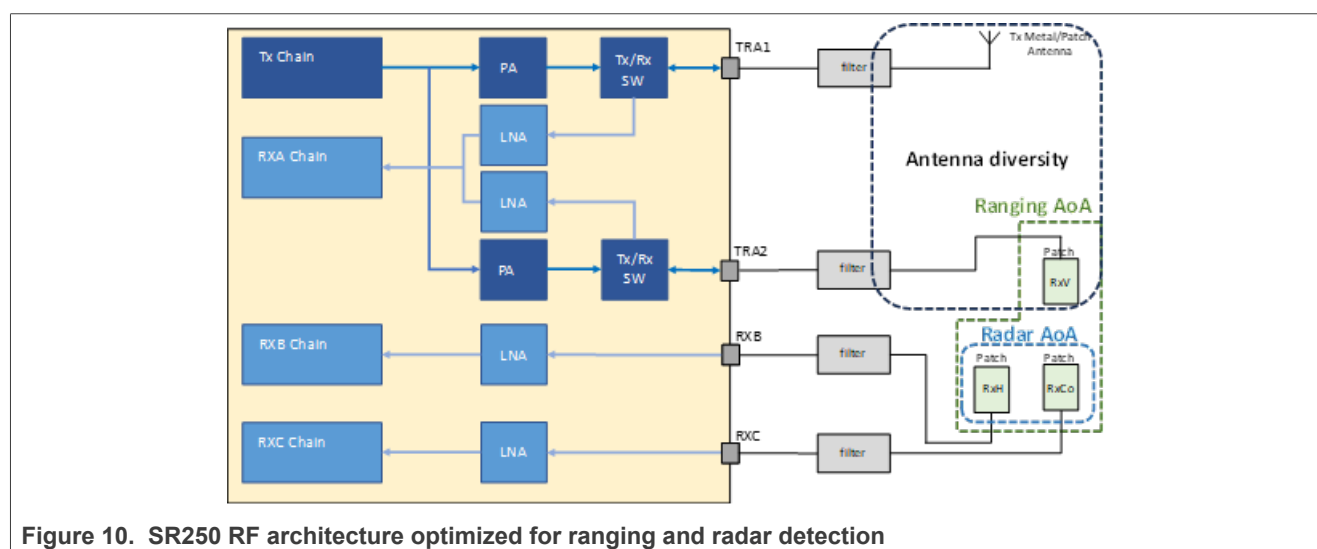


Figure 10. SR250 RF architecture optimized for ranging and radar detection

6.1.6 TX-RX isolation requirements (radar)

As radar is running full duplex (for example, TX and RX are running at the same time), TX signal leakage to the receiver chain must be minimized in order to avoid RX path saturation and reach maximum radar distance performance.

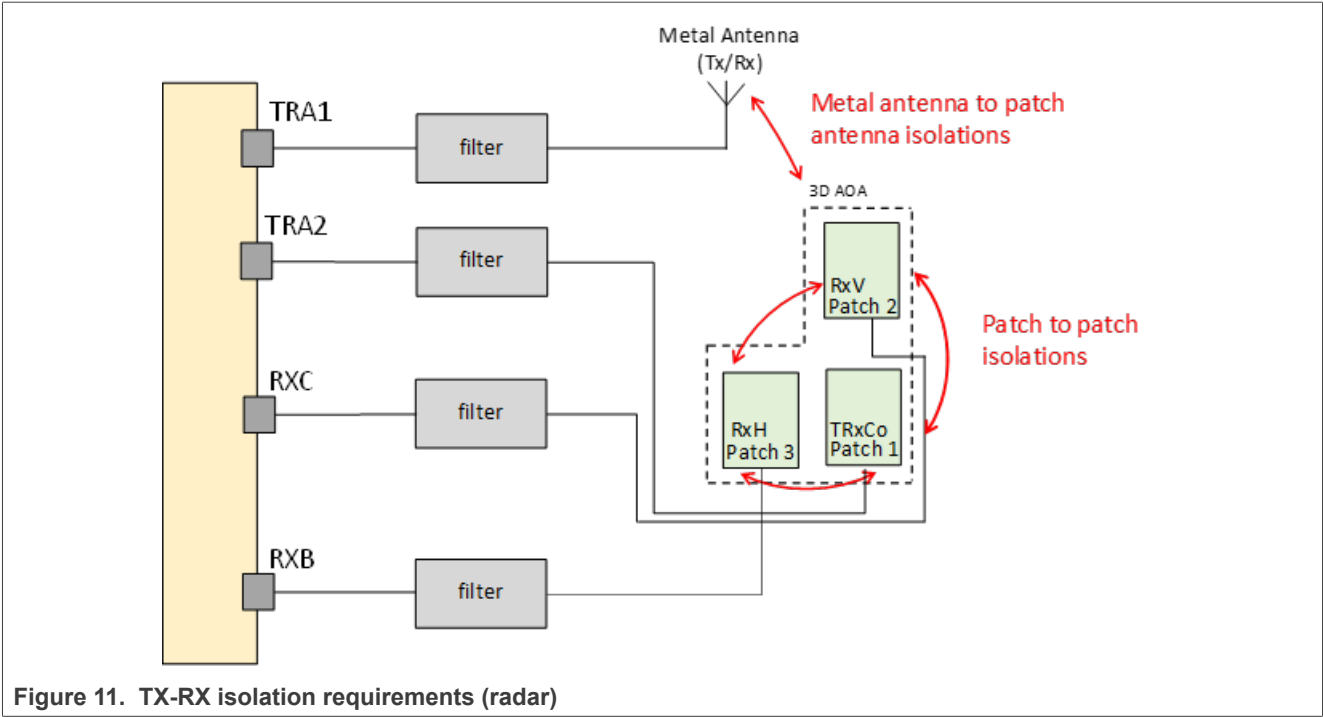
For the SR250, the maximum radar distance achieved is limited by the isolation between the TX antenna and the RX antenna in use.

A typical 35 dB to 40 dB isolation range offers optimal radar performances in terms of maximum ranging distance.

Having lower isolation will have an impact on the sensitivity (dB-by-dB degradation), in turn, reducing the maximum radar ranging distance.

A minimum isolation of approximately 20dB is recommended as a limit to achieve a maximum radar detection distance of about 1m.

In a configuration as illustrated in Figure 11, optimum radar distance performance can be achieved when the transmitter is connected to the metal antenna and the receiver is connected to the patch antenna. In typical UWB mobile integration, patch-to-patch isolation is in a range of 15 dB to 20dB.



6.2 RF paths description: BOM

6.2.1 RF filter

The chip-referred blocker level is a key parameter, and results predominantly out of filter rejection and antenna isolation. Therefore, the UWB band-pass filters must remove the unwanted signals and noise in the UWB communication system. They must provide high selectivity, harmonic suppression, and broad rejection bandwidth.

On each RF path a ceramic band-pass filter is placed in order to pass the UWB channels 5 and 9 and provide rejection of co-existing bands such as cellular, LTE, Bluetooth, and WiFi.

Murata’s **LFB217G35CFHE826** is designed to provide rejection from 698 MHz up to 5.725 GHz (up to U-NII-2e).

If rejection of U-NII-3 and U-NII-4 is required, NXP has developed the **LFB217G49CFHF022** with Murata to provide rejection from 698 MHz up to 5.925 GHz, at the expense of some passband roll-off in Channel 5.

The recommended Murata filter references and characteristics are listed below:

Table 9. Murata band-pass filter rejection characteristics

Freq range (MHz)	LFB217G35CFHE826	LFB217G42CFHV942	LFB217G24CFKF265	LFB187G87CTPF642
698 ~ 2200	36.2 dB typ.	28.6 dB typ.	35.0 dB min.	-
2300 ~ 3800	35.0 dB typ.	28.6 dB typ.	35.0 dB min.	-
5150 ~ 5470	24.5 dB typ.	18.8 dB typ.	15.0 dB min.	20.7dB typ.
5470 ~ 5725	13.4 dB typ.	18.8 dB typ.	15.0 dB min.	20.7dB typ.
5725 ~ 5850	6.1 dB typ.	18.8 dB typ.	15.0 dB min.	20.7dB typ.
5850 ~ 5925	3.7 dB typ.	11.1 dB typ.	10.0 dB min.	12.1 dB typ.

Table 9. Murata band-pass filter rejection characteristics...continued

Freq range (MHz)	LFB217G35 CFHE826	LFB217G42CFHV942	LFB217G24CFKF265	LFB187G87 CTPF642
9200 ~ 12500	-	22.9 dB typ.	-	-
12500 ~ 16500	23.7 dB typ.	22.9 dB typ.	20.0 dB min.	24.7 dB typ.
18700 ~ 20000	13.1 dB typ.	22.9 dB typ.	20.0 dB min.	24.7 dB typ.

6.2.2 RF connectors

It is strongly recommended to add an RF connector to be able to spy the RF signal on the TX path or to inject an RF Signal toward RX inputs for debugging and calibration purposes.

Typically, MS-156C3 is well suited for this functionality.

Moreover, adding the possibility to route the RF path to a U.FL connectors makes it possible to use external antennas, such as Taoglas Flex antenna.

6.2.3 Routing of RF paths and layout recommendations

This section provides layout recommendations for the RF paths. To illustrate the guidelines, a snapshot of the RF part of the Shield board layout is shown in [Figure 13](#), while the electrical schematic is provided in [Figure 12](#) as a simplified block diagram.

- RF paths must be perfectly symmetrical in terms of the components used, their placement, and the copper pads and shapes (transmission lines) joining them.
- RF traces must be 50 Ω and components should be matched to 50 Ω to minimize mismatch losses.
- The transmission line and reference plane layers must be at stitched thoroughly with ground vias.
- Maintain good isolation between all RF paths.
- The antenna structure should follow the manufacturer's guidelines with regard to position with respect to the ground plane and the printed-circuit board, thickness of the PCB and dielectric constant. Where these are driven by other factors, adjustments to the antenna structure may be necessary.
- No component or ground plane should be placed in the antenna area.
- UWB radio **must** be shielded to prevent unwanted emissions and to provide immunity from external sources of interference.

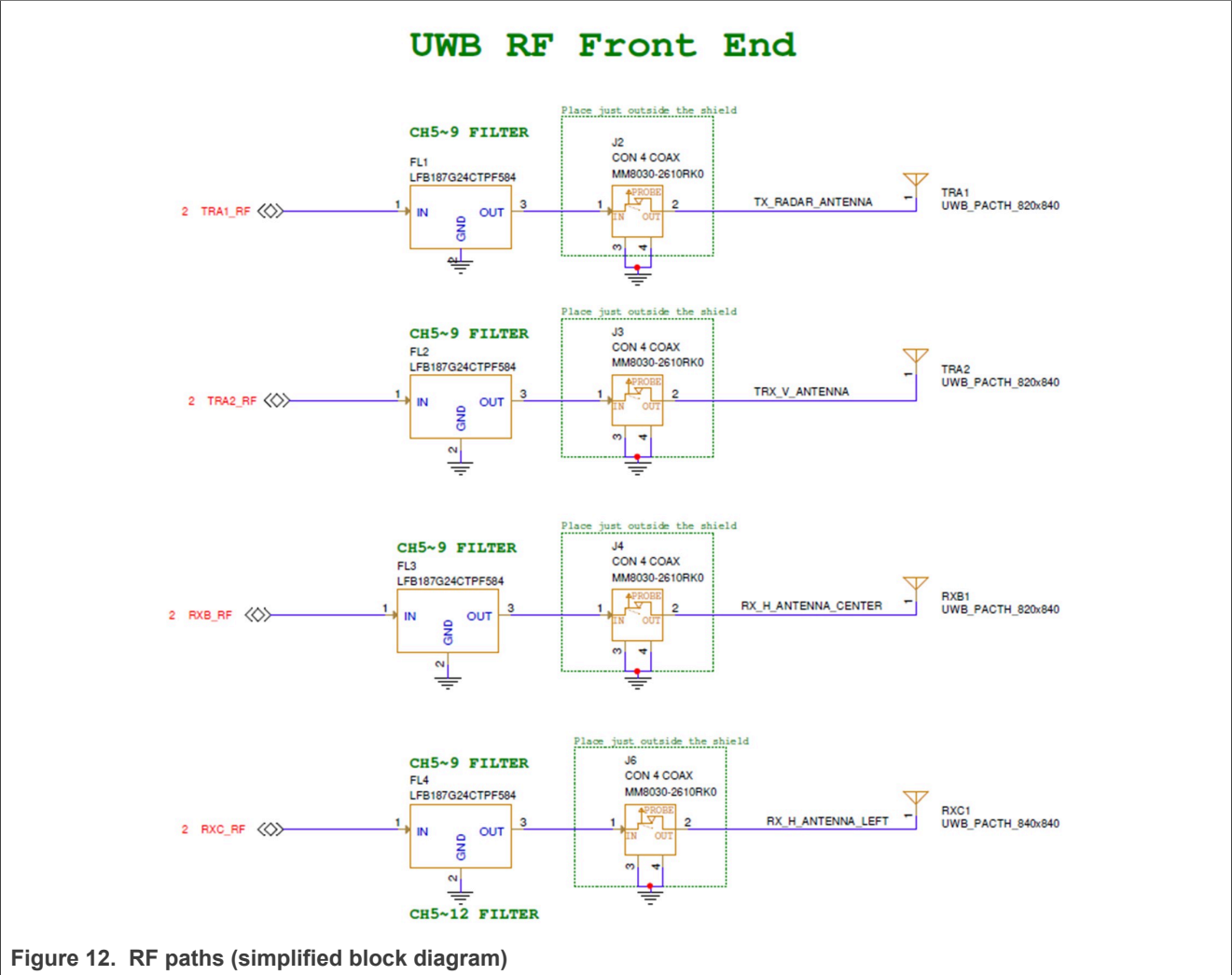


Figure 12. RF paths (simplified block diagram)

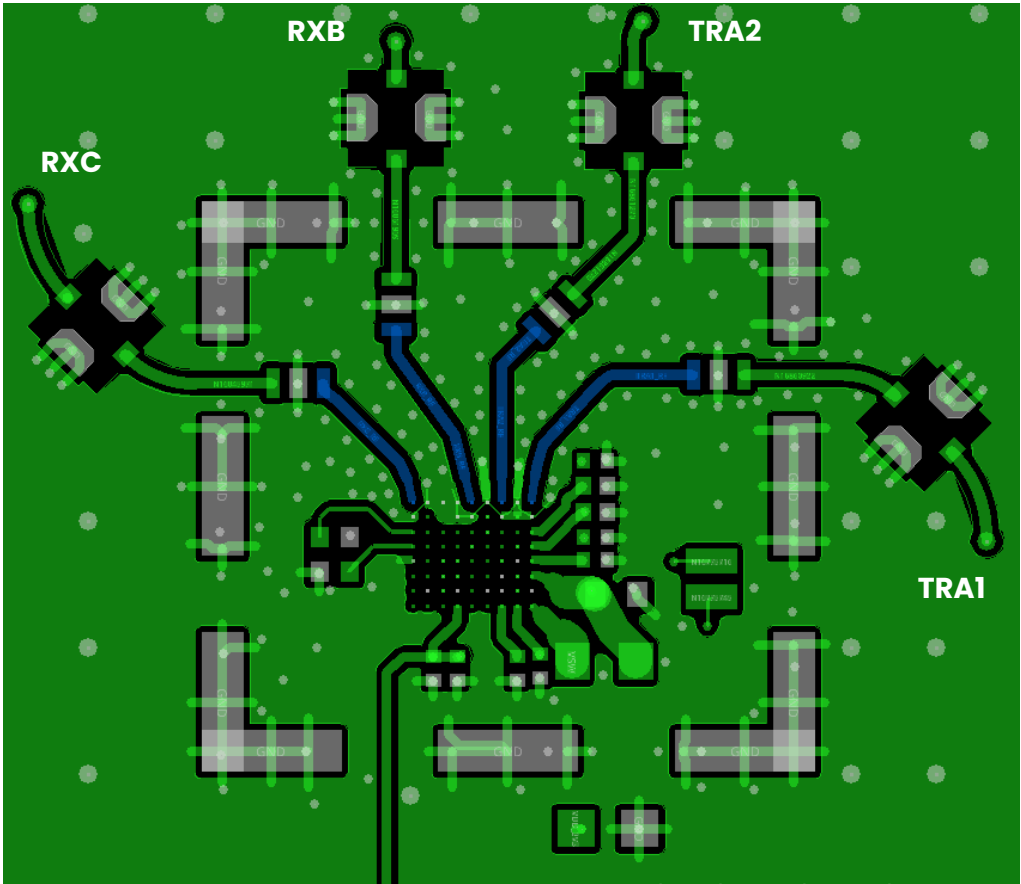


Figure 13. RF paths Layout example

7 Placement and Layout recommendations

The placement and layout recommendations are presented in the following sections.

Typically, six or more layers are required to form a full Faraday cage around all active traces.

For recommendations on specific aspects of the application design, refer to the relevant sections:

- [Section 7.1 "Routing layers"](#)
- [Section 7.2 "Ground aspects"](#)
- [Section 7.3 "Component placement"](#)
- [Section 7.4 "DC-DC buck converter pins"](#)
- [Section 7.5 "RF layout"](#)
- [Section 7.6 "Shielding"](#)

7.1 Routing layers

- Minimum four routing layers required to escape from the center pins, though six or more layers are recommended for proper EMI design.
- A High Density Interconnections (HDI) microvia stack from L1 to L4 is preferable.
- Possible with a L3-L4 buried core via but less ideal as this typically needs to be staggered.
- 7.874 mil (178 μm) L1 pad for soldering
- 10 mil (250 μm) inner layer pad
- 4 mil (100 μm) HDI laser-drilled and filled via

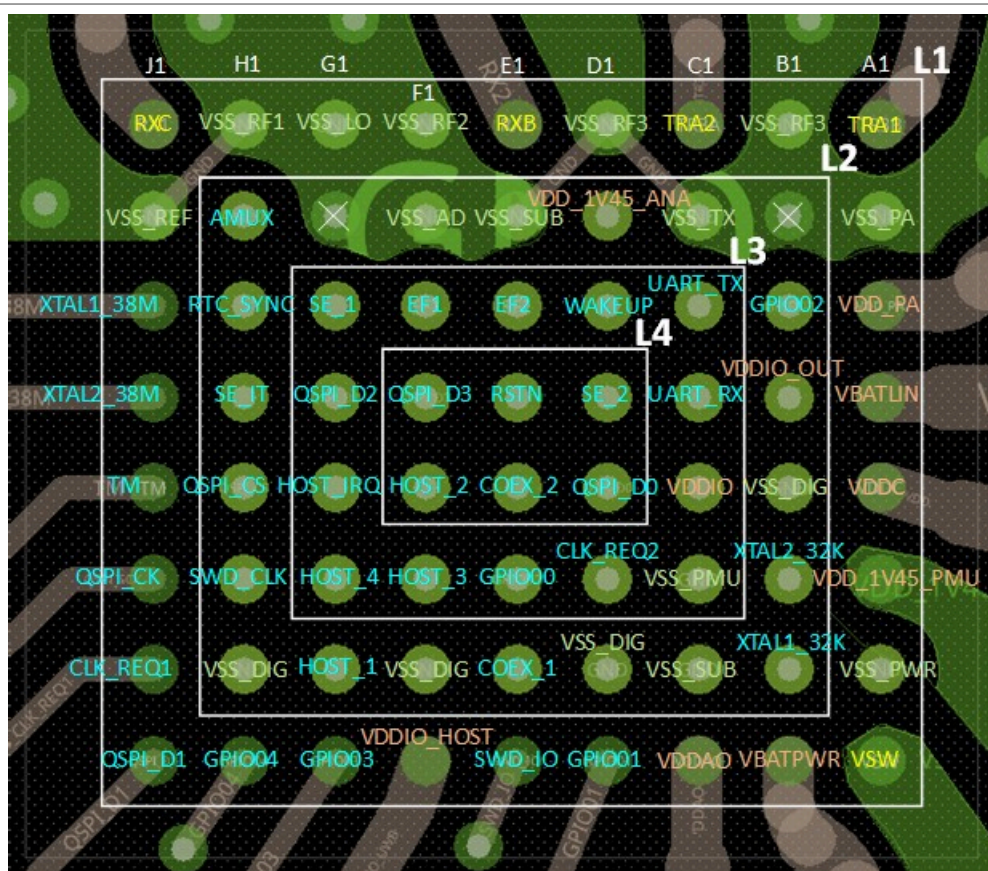


Figure 14. Ball row vs routing layer

[Figure 15](#) to [Figure 19](#) illustrate how the IC ball rows are routed according to the PCB layers.

• Layer 1:

- Consists of the RF pins and grounding. RF pins are directly routed using 50 Ω tracks, with the reference ground taken on L3 to keep track width wide enough to limit path losses.
- It also supports:
 - power supply lines
 - XTAL_38M
 - peripheral control signals

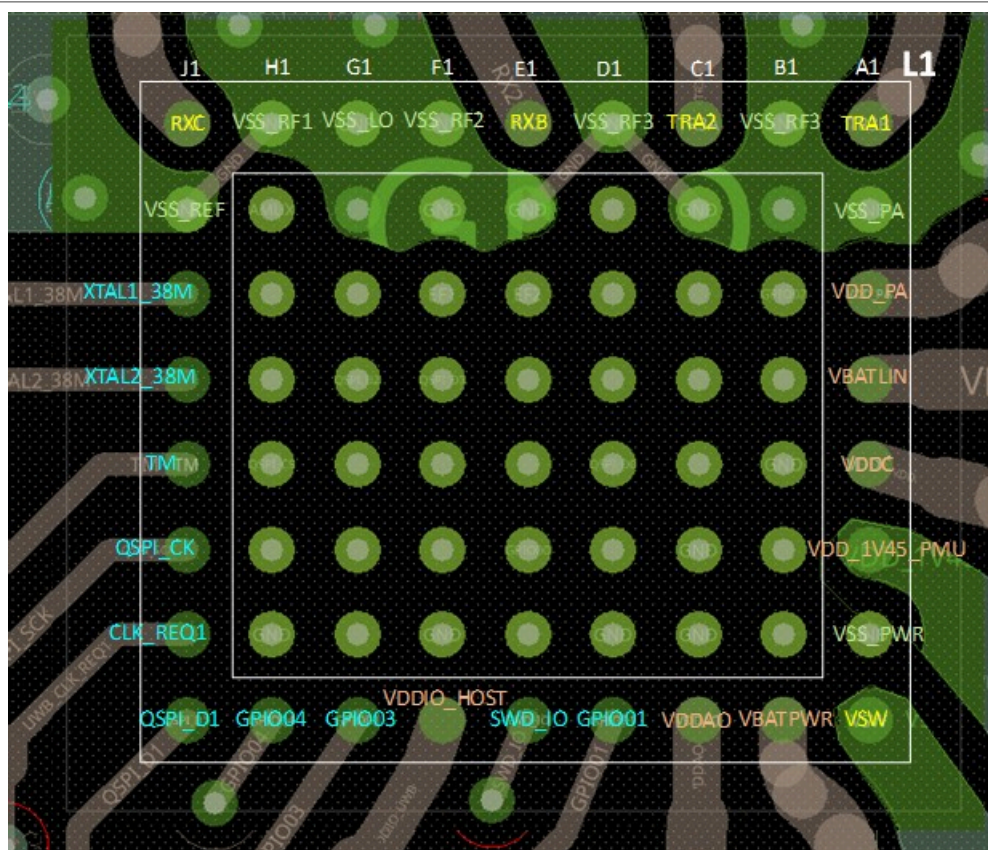


Figure 15. Ball row vs routing layer (L1)

- **Layer 2:**
 - Supports ground plane.
 - Clearances under RF tracks can be observed.
 - Active balls are connected to L2 through L1-L2 and L2-L3 vias.

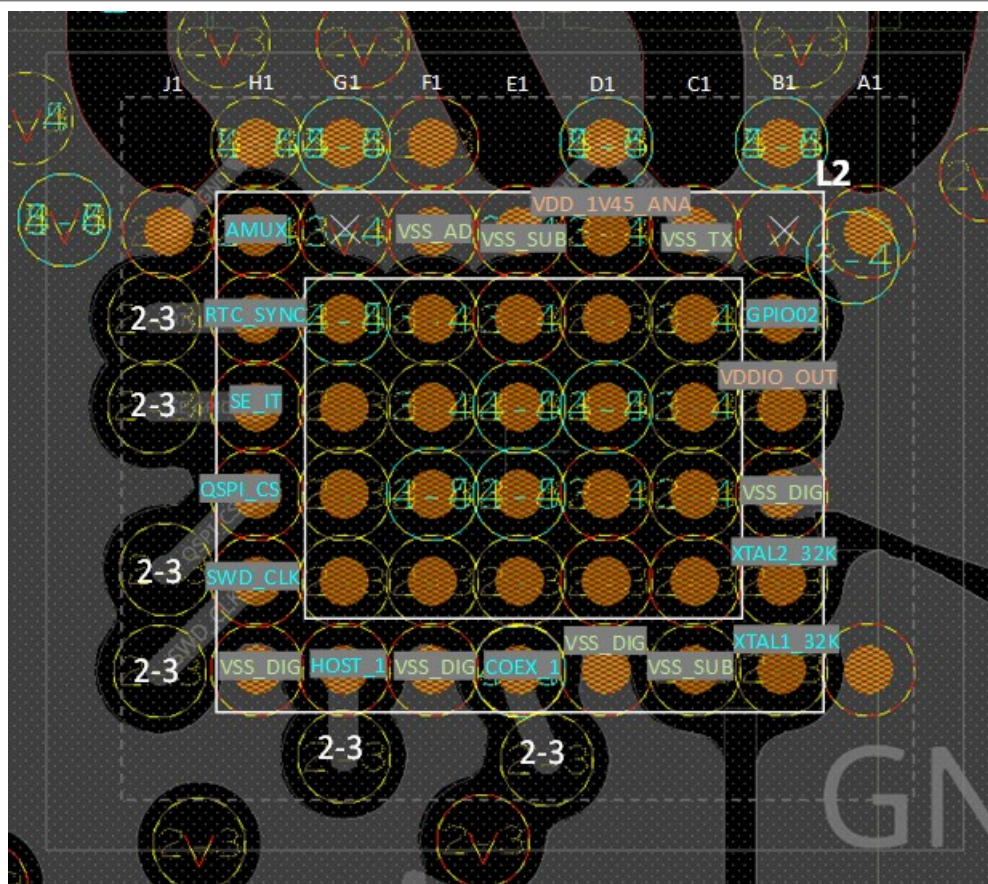


Figure 16. Ball row vs routing layer (L2)

- **Layer 3:**

Supports:

- ground plane
- control signals
- VDDIO OUT
- XTAL 32K

Keep VSS_PMU(C6) and VSS_SUB(C7) pins away from the noisy neighboring grounds on the VSS_DIG pins (B5, D7).

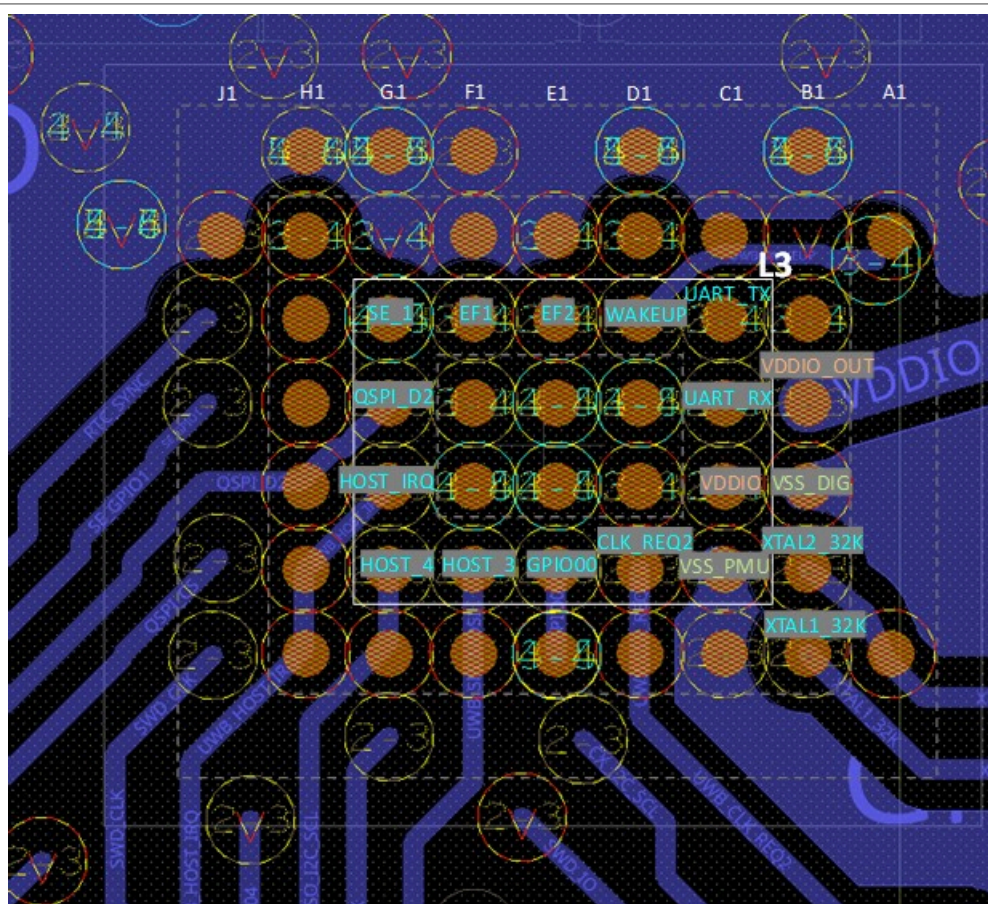


Figure 17. Ball row vs routing layer (L3)

• Layers L4-L5:

Supports:

- power supply ground plane
- power supplies VDD_1V45_ANA and VDDIO
- remaining control signals

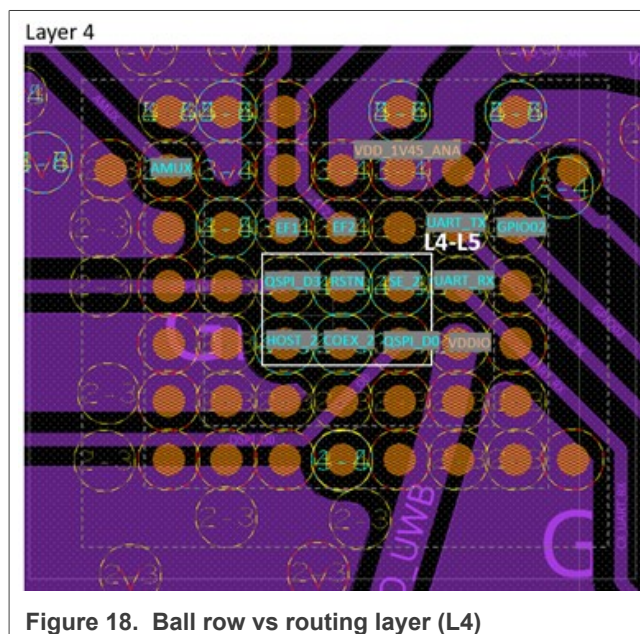


Figure 18. Ball row vs routing layer (L4)

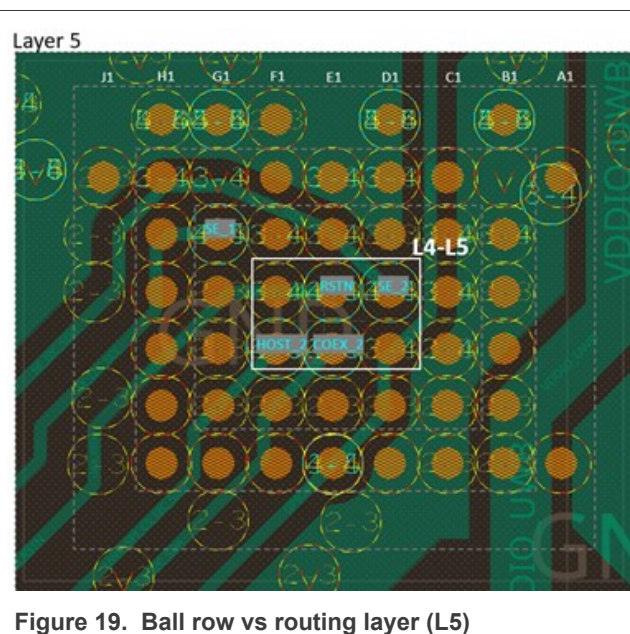


Figure 19. Ball row vs routing layer (L5)

• Layers L6-L7:

Supports:

- full ground planes

7.2 Ground aspects

7.2.1 Ground plane

- One layer is required for solid ground. Since L4 is not completely solid, use two more layers to support power delivery on L5 and a solid ground on L6.
- All shunt components have a via-in-pad direct connection to the GND plane for minimal ground inductance.

Ground fill:

- Flood all other routing layers with GND plane.
- Stitch them together well with GND vias.
- Note that since the ground plane is broken up by the traces (on L1 to L5), the solid ground plane is required on L6.
- Where possible, SR250 ground pins can also connect to these layers.

7.2.2 SR250 ground pins

- RF ground:
 - RF ground is stacked over L1-L2-L3
 - Clearance under RF lines is done on L2.
 - RF GND pins via-in-pad connection to the microstrip reference plane (L3 in this case)
- Direct via-in-pad connection to GND plane (L6+L7 in this case) to minimize ground inductance.
- Where via-in-pad is not possible a short GND trace is acceptable

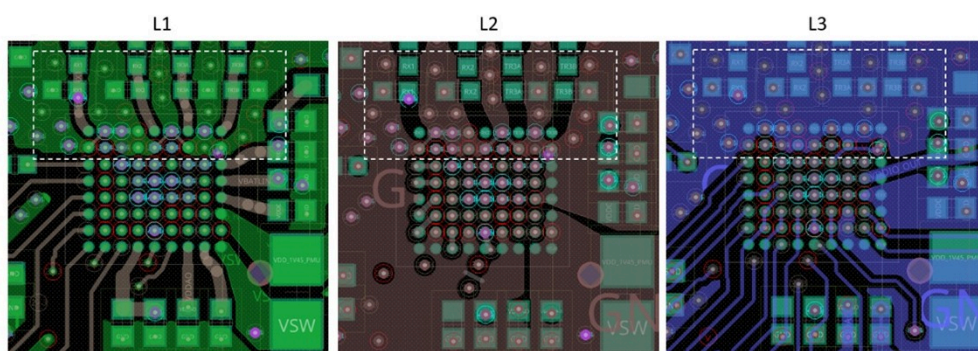


Figure 20. SR250 RF GND pins

7.3 Component placement

7.3.1 VDD LDO regulator outputs

There are three LDO regulator output pins: VDD_PA (A3), VDDC (A5), VDDAO (C8) that require a close decoupling capacitor.

- To minimize VDD trace inductance, keep L1 VDD inductances on L1 under 0.5 nH and inductances in inner layers under 1.0 nH
- Trace width: 10 mil minimum (250µm).

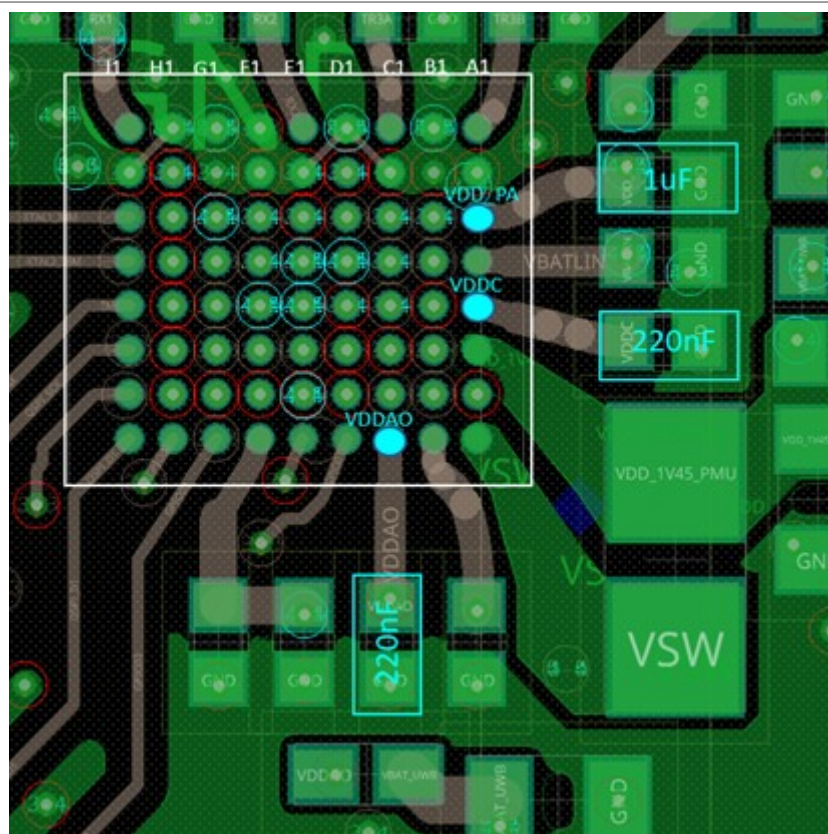


Figure 21. LDO decoupling capacitor placement

7.3.2 38.4 MHz crystal

- Keep XTAL1_38M and XTAL2_38M short and separated from one another.
- Trace width: 5 mil (127 µm)
- Minimize stray capacitance under the active pads by clearing nearby ground (in this case, L1 through L4)
- Keep stray C on each trace below 0.5 pF
- Recommend to route the 38.4 MHz crystal traces on L1
- Keep good separation from and do not cross any other active traces (additional load capacitors are not required if an 8 pF XTAL is selected)

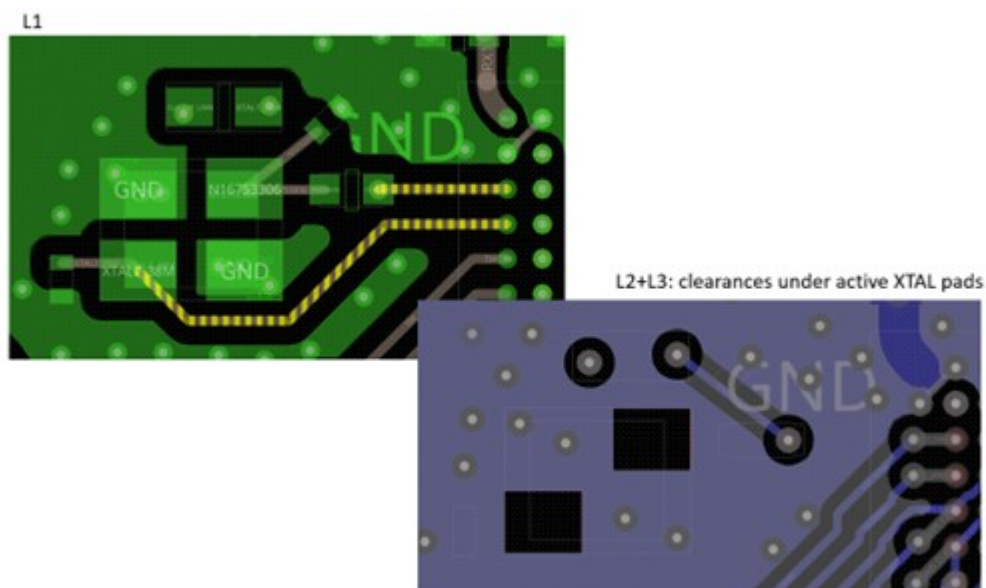


Figure 22. 38.4 MHz crystal layout

7.3.3 32.768 kHz crystal

- Keep XTAL1_32K and XTAL2_32K short and separated from one another.
- Trace width: 5mil (127 μ m)
- Minimize stray capacitance under the active pads by clearing nearby ground (in this case, L1 through L4)
- Keep stray C on each trace below 1 pF.
- 32.768 kHz crystal traces can be routed on inner layers.
- Keep good separation from and do not cross any other active traces (additional load capacitors are not required if a 4 pF XTAL is selected).

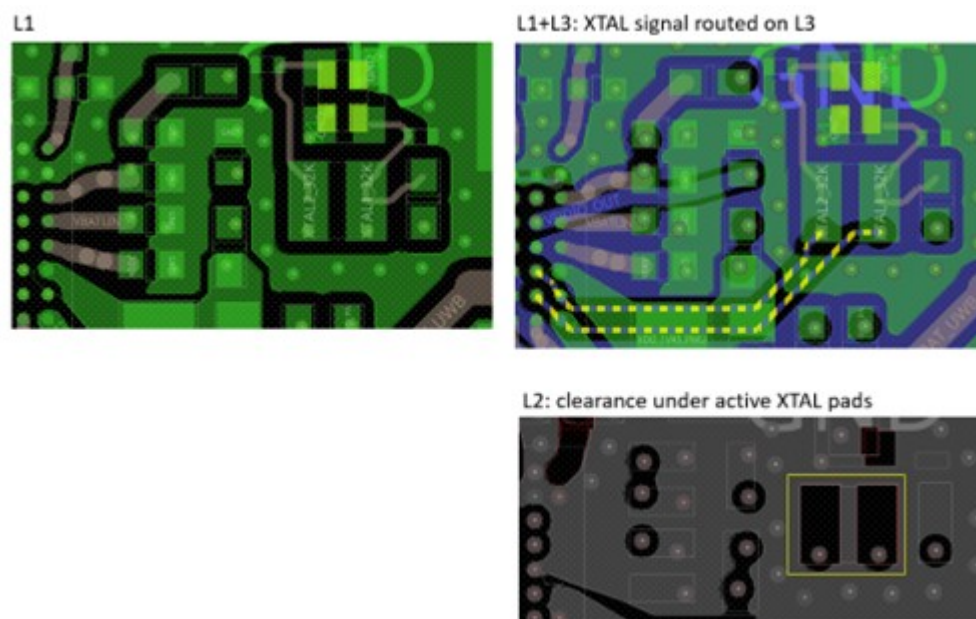


Figure 23. 32.768 kHz crystal layout

7.3.4 VDD power supply inputs (1.8 V)

There are three main power supply inputs, VBATPWR, VBATLIN, and VDDIO.

- Place the smallest decoupling capacitor (1 uF, 0.22 uF) as close as possible to the associated pin.
- Place larger decoupling capacitor (15 uF) in the second row.
- Keep VDD inductances on L1 under 0.5 nH and inductances in inner layers under 1.0 nH.
- Trace width: 10 mil minimum (250µm)

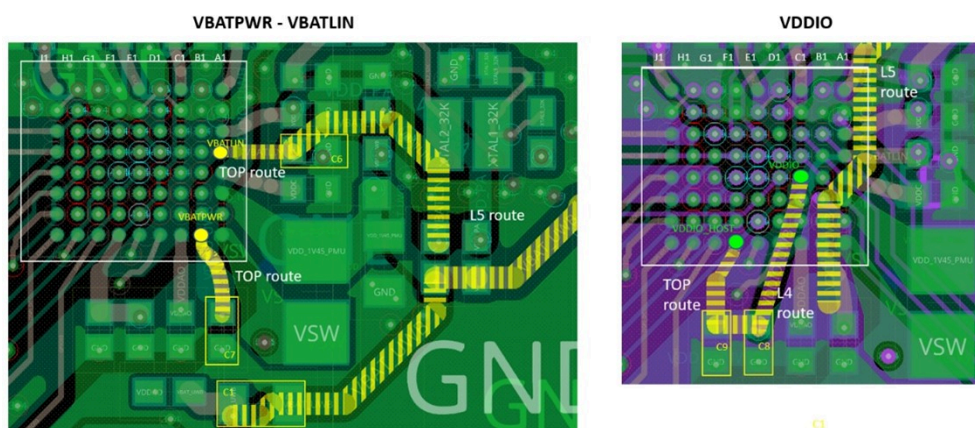


Figure 24. VBAT Power supplies decoupling capacitor placement and power rail distribution

7.4 DC-DC buck converter pins

The DC-DC buck has three pins, VDD_1V45_PMU, VSW, and VDD_1V45_ANA.

- The DC-DC inductor must be placed as close as possible to the pins with large routes on the top layer. Attach the decoupling capacitor in the second row to the inductor.
- The VDD_1V45_ANA decoupling capacitor can be routed on L4
- Keep inductances in the inner layers below 1.0 nH.
- Trace width: 10mil minimum (250µm)

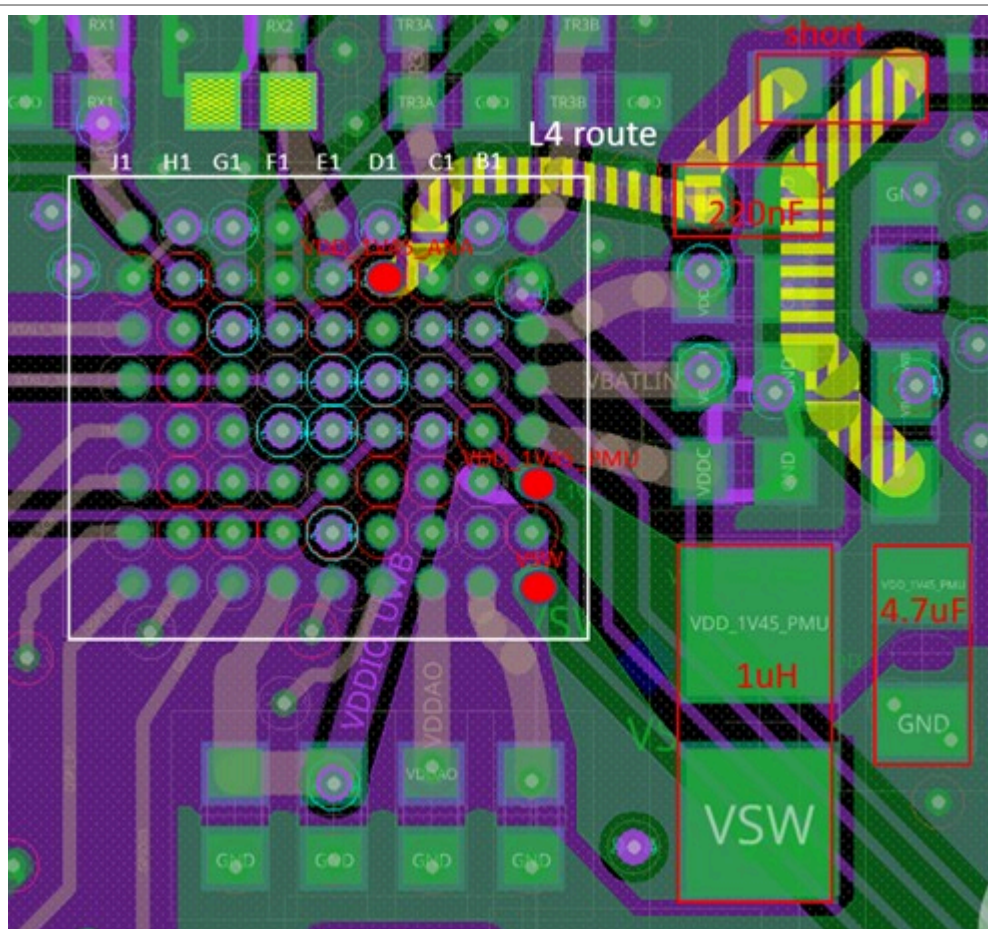


Figure 25. DC-DC inductor and decoupling capacitor placement

7.5 RF layout

- To minimize RF trace loss and match CPW width with to 0201 or 01005 pad width, clear at least 1 HDI layer of ground (in this case L2 is cleared)
- In this case, the CPW trace is 11.5 mil and references L3. Gap to L3 is 10.64 mil.
- Stitch L1, L2 and L3 (reference layer) with GND vias at most every 40 mil (1 mm)
- PCB fabricator must agree on 50 Ω CPW dimensions and FAB DWG must include 50 Ω controlled impedance requirements.

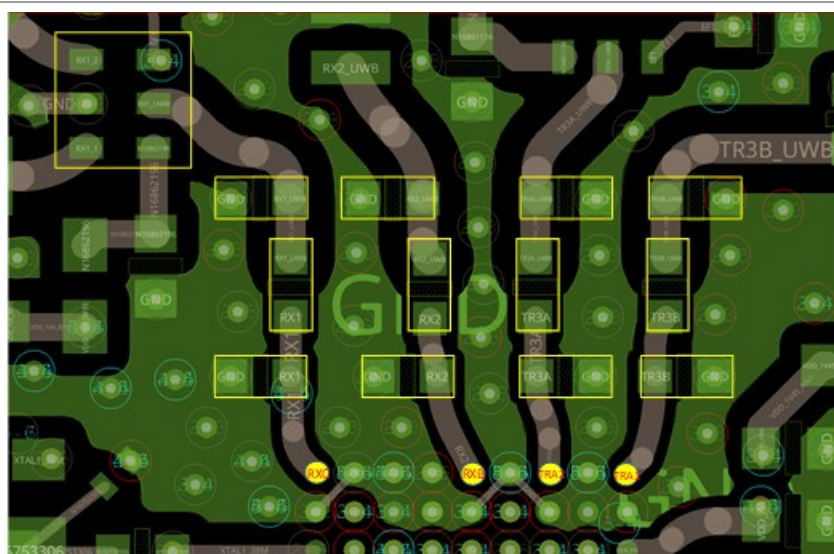


Figure 26. RF trace layout

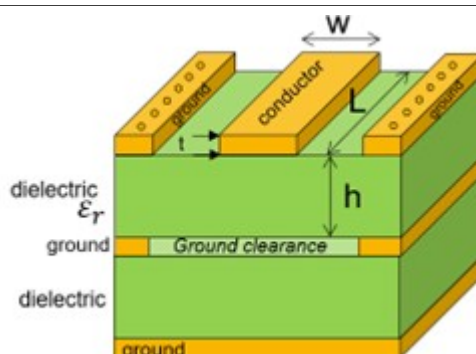


Figure 27. Standard microstrip RF trace design

7.6 Shielding

The SR250 and close hardware (i.e., decoupling caps, filters, XTALs, ...) **must** be entirely shielded in order to limit radiated frequency spurs that could lead to issues during board certification.

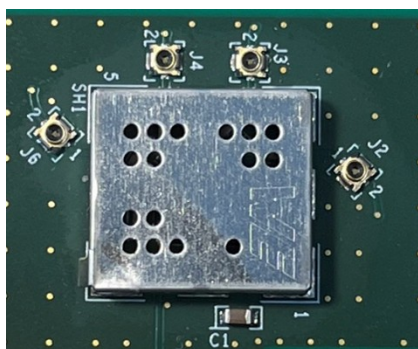


Figure 28. Shielding (1)

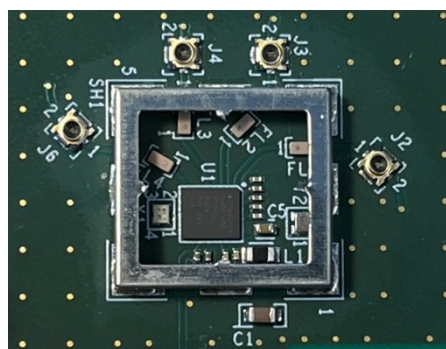


Figure 29. Shielding (2)

8 Abbreviations

Abbr.	Meaning
AN	Application note
AOA	Angle Of Arrival
BoM	Bill of materials
CLK	Clock
DUT	Device Under Test
DPD	Deep Power Down
ETSI	European Telecommunications Standards Institute
FoV	Field of View
GND	Ground
GPIO	General Purpose Input Output
HPD	Hard Power Down
HW	Hardware
IC	Integrated Circuit
IO	Input / Output
IRQ	Interrupt Request
mA	milliampere
MHz	megahertz
mW	Milliwatt
PDoA	Phase Difference of Arrival
PMU	Power Management unit
RF	Radio Frequency
SE	Secure Element (smart card controller)
SPI	Serial Peripheral Interface
UWB	Ultra-Wideband
XTAL	Crystal

9 References

- [1] Application note – AN13848 – SR250 calibration procedure guide ([link](#))
- [2] User manual – UM12508 – SR250 UCI specification ([link](#))

10 Revision history

Table 10. Revision history

Document ID	Release date	Description
AN14875 v.1.0	27 March 2026	Initial version

Legal information

Definitions

Draft — A draft status on a document indicates that the content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included in a draft version of a document and shall have no liability for the consequences of use of such information.

Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use — NXP Semiconductors products are not designed, authorized or warranted to be suitable for use in life support, life-critical or safety-critical systems or equipment, nor in applications where failure or malfunction of an NXP Semiconductors product can reasonably be expected to result in personal injury, death or severe property or environmental damage. NXP Semiconductors and its suppliers accept no liability for inclusion and/or use of NXP Semiconductors products in such equipment or applications and therefore such inclusion and/or use is at the customer's own risk.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale — NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at <https://www.nxp.com/profile/terms>, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Suitability for use in non-automotive qualified products — Unless this document expressly states that this specific NXP Semiconductors product is automotive qualified, the product is not suitable for automotive use. It is neither qualified nor tested in accordance with automotive testing or application requirements. NXP Semiconductors accepts no liability for inclusion and/or use of non-automotive qualified products in automotive equipment or applications.

In the event that customer uses the product for design-in and use in automotive applications to automotive specifications and standards, customer (a) shall use the product without NXP Semiconductors' warranty of the product for such automotive applications, use and specifications, and (b) whenever customer uses the product for automotive applications beyond NXP Semiconductors' specifications such use shall be solely at customer's own risk, and (c) customer fully indemnifies NXP Semiconductors for any liability, damages or failed product claims resulting from customer design and use of the product for automotive applications beyond NXP Semiconductors' standard warranty and NXP Semiconductors' product specifications.

HTML publications — An HTML version, if available, of this document is provided as a courtesy. Definitive information is contained in the applicable document in PDF format. If there is a discrepancy between the HTML document and the PDF document, the PDF document has priority.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at PSIRT@nxp.com) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

NXP B.V. — NXP B.V. is not an operating company and it does not distribute or sell products.

Trademarks

Notice: All referenced brands, product names, service names, and trademarks are the property of their respective owners.

NXP — wordmark and logo are trademarks of NXP B.V.

Tables

Tab. 1.	External power supplies voltages	5	Tab. 6.	SPI direction bytes decoding	7
Tab. 2.	Decoupling capacitors (default configuration)	5	Tab. 7.	XTAL references for high frequency clock	8
Tab. 3.	External power supplies voltages	6	Tab. 8.	XTAL references for low frequency clock	8
Tab. 4.	Decoupling capacitors (BOM optimized configuration)	6	Tab. 9.	Murata band-pass filter rejection characteristics	13
Tab. 5.	Host connections	7	Tab. 10.	Revision history	32

Figures

Fig. 1.	SR250 logic block diagram	2	Fig. 13.	RF paths Layout example	16
Fig. 2.	SR250 reference application electrical schematic – Default configuration	3	Fig. 14.	Ball row vs routing layer	18
Fig. 3.	SR250 reference application electrical schematic – BOM optimized	4	Fig. 15.	Ball row vs routing layer (L1)	19
Fig. 4.	SR250 host interface signals	7	Fig. 16.	Ball row vs routing layer (L2)	20
Fig. 5.	SR250 single antenna reference schematic – RF antenna path	9	Fig. 17.	Ball row vs routing layer (L3)	21
Fig. 6.	SR250 RF architecture optimized for UWB ranging ToF	10	Fig. 18.	Ball row vs routing layer (L4)	22
Fig. 7.	SR250 RF architecture optimized for ranging AoA	10	Fig. 19.	Ball row vs routing layer (L5)	22
Fig. 8.	SR250 RF architecture optimized for radar ToF	11	Fig. 20.	SR250 RF GND pins	23
Fig. 9.	SR250 RF architecture optimized for radar AoA	11	Fig. 21.	LDO decoupling capacitor placement	24
Fig. 10.	SR250 RF architecture optimized for ranging and radar detection	12	Fig. 22.	38.4 MHz crystal layout	25
Fig. 11.	TX-RX isolation requirements (radar)	13	Fig. 23.	32.768 kHz crystal layout	25
Fig. 12.	RF paths (simplified block diagram)	15	Fig. 24.	VBAT Power supplies decoupling capacitor placement and power rail distribution	26
			Fig. 25.	DC-DC inductor and decoupling capacitor placement	27
			Fig. 26.	RF trace layout	28
			Fig. 27.	Standard microstrip RF trace design	28
			Fig. 28.	Shielding (1)	29
			Fig. 29.	Shielding (2)	29

Contents

1	Introduction	2
2	Typical application schematics	3
2.1	SR250 reference schematic – Default configuration	3
2.2	SR250 reference schematic – BOM optimized	3
3	Power interface	5
3.1	Default configuration	5
3.1.1	External power supplies	5
3.1.2	Decoupling capacitors (default configuration)	5
3.2	BOM optimized configuration	6
3.2.1	External power supplies	6
3.2.2	Decoupling capacitors (BOM optimized configuration)	6
4	Host interface (SPI)	7
5	CLK interface (XTAL)	8
5.1	High frequency clock XTAL	8
5.2	Low frequency clock XTAL	8
6	RF architecture	9
6.1	RF architecture: UWB ranging and radar	9
6.1.1	UWB ranging ToF	9
6.1.1.1	Ranging basic ToF	9
6.1.1.2	Ranging ToF with antenna diversity	9
6.1.2	UWB ranging AoA	10
6.1.3	UWB radar ToF	10
6.1.4	UWB radar detection AoA	11
6.1.5	Optimized architecture for UWB ranging and radar detection (ToF/AoA)	12
6.1.6	TX-RX isolation requirements (radar)	12
6.2	RF paths description: BOM	13
6.2.1	RF filter	13
6.2.2	RF connectors	14
6.2.3	Routing of RF paths and layout recommendations	14
7	Placement and Layout recommendations	17
7.1	Routing layers	18
7.2	Ground aspects	23
7.2.1	Ground plane	23
7.2.2	SR250 ground pins	23
7.3	Component placement	24
7.3.1	VDD LDO regulator outputs	24
7.3.2	38.4 MHz crystal	24
7.3.3	32.768 kHz crystal	25
7.3.4	VDD power supply inputs (1.8 V)	26
7.4	DC-DC buck converter pins	27
7.5	RF layout	28
7.6	Shielding	29
8	Abbreviations	30
9	References	31
10	Revision history	32
	Legal information	33

Please be aware that important notices concerning this document and the product(s) described herein, have been included in section 'Legal information'.